INFOSCI 301 FINAL GROUP PROJECT POSTER

Project Title: A Source-Linked Visualization System for Comparing Recorded Color Patterns Across Places

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Instructor: Prof. Luyao Zhang

Project frame and evidence

Question How can users compare colors recorded in different places and understand where those colors come from?

Intended user Design and media students, creator, and cultural educators.

Exact task Select two places, compare their recorded color patterns, and trace one similarity or difference back to scene categories and source photographs.

Local context Jinxi field observation and cross-place photo comparison revealed recurring visual patterns worth investigating, but whether these patterns are locally meaningful or broadly representative remains unvalidated.

Claim The current prototype supports comparison of recorded color patterns across five photographic samples and allows users to trace aggregate patterns back to individual source photographs.

Evidence boundary Current samples support comparison, but not representative claims; results may reflect sampling, lighting, and photographer choice.

Five representations, one task

Basic	Palette ranking and A/B comparison show recorded color patterns in the current photo samples.
Network	Places are nodes; edges show sampled palette similarity.
Spatial or temporal	The geographic atlas connects palettes and source photos to place context.
Interactive	Users browse, compare, and trace aggregate patterns back to source photographs.
Advanced	A draggable photo divider updates the visible blend and weighted blend palette .

Verified data and provenance **Producer**: Project members
URL: github.com/charleston1107/suzhou-in-color
Access: Team-authored; public repository, no open-source license
Coverage: 50 photographs across 5 places
Missingness: Small, uneven samples; local people researching do not complete and photographer bias remain.

Repository and software architecture

1 Data and provenance
50 team-authored photos from five places with source-linked metadata.

→ **2 Transform and store**
Photos are normalized and reduced to five dominant CIELAB colors with deterministic K-means.

→ **3 Logic and algorithms**
CIEDE2000 palette matching and proportion weighting produce 0–100 similarity scores.

→ **4 Views and interaction**
Geographic atlas, source views, A/B comparison, similarity network, and Compare slider.

→ **5 Deploy, test, fallback**
GitHub Pages deployment with reproducibility checks and local static-server fallback.

Repository URL https://github.com/charleston1107/atlas-of-recorded-color **Version or commit** 754758f
Artifact URL https://charleston1107.github.io/atlas-of-recorded-color/

Input: 50 source photographs with place, category, and photo metadata **Transformation:** CIELAB and deterministic K-means++ dominant-color extraction **Logic:** CIEDE2000 palette matching with proportion weighting **Output:** geographic atlas, A/B comparison, similarity network, and Compare interaction **Checks:** reproducibility, self-similarity, symmetry, and browser/offline consistency **Fallback:** documented local static-server workflow

1 User and task
Design/media peer explores place palettes, compares samples, checks similarity, and traces results to source photos.

→ **2 Starting state**
Default geographic atlas/homepage with five sampled places.

→ **3 User action**
Select a place or photo, compare samples, inspect palette/similarity, and use the Compare interaction.

→ **4 System response**
Shows source-linked palettes, similarity results, source images, and the weighted blend palette.

→ **5 Observed evidence**
Classroom critique found weak algorithm explanation, missing advanced effect, and visibility issues.

→ **6 Decision and revision**
Revised the algorithm, added the advanced Compare interaction, improved visibility, and kept temporal comparison as Investigate.

Participant role and notice In-class design/media peers and instructor **Sep 29 evidence** About 2 verbal and 5 written comments on algorithm grounding, advanced effect, scope, and readability. **What remains untested** Broad usability, learning, cultural validity, community benefit, and SDG impact. **Next test and owner** Broader intended-user testing and algorithm–human similarity validation. Owners: Interaction & Evaluation; Data Visualization & Prototyping



Before: Early geographic map view with unclear visual structure.



After: Simplified schematic map with clearer place and palette reading.

Four validation levels

Domain situation Evidence: Jinxi fieldwork and cross-place comparison support a place-based color comparison question.**Main threat**: Local relevance remain unvalidated.

Data and task Evidence: Five locations and source-linked photographs support Select to Compare to Trace.**Main threat**: Small, uneven samples may overrepresent route, weather, lighting, or photographer choice.

Idiom Evidence: Ranking, network, and photo detail support compare/discover/trace.**Main threat**: Similarity may be overread as cultural similarity.

Algorithm Evidence:K-means + CIELAB + ΔE00 support perceptual palette comparison.**Main threat**: Rankings may change with clustering, weighting, or parameter choices.

Interdisciplinary contribution

Open science, data governance, and responsible AI
Reproducible, source-linked data and documented AI use improve traceability; reuse and cultural-context review remain limited.

Scientific communication for emerging technologies
Peer feedback led to clearer methods, evidence limits, and source-linked results; validation is still based on class.

Human-centered spatiotemporal intelligence
Geographic atlas links palettes to place and source photos; temporal context and local validation remain limited.

Community-rooted global leadership

Local community or audience DKU design/media students, creative practitioners, and cultural educators

Service and accountability Missing voices include residents and cultural practitioners; source transparency and a future correction route support accountability.

SDG goal and exact target SDG 11.4: document and communicate sampled place-based visual material with provenance; alignment only, not demonstrated heritage impact.

CURRENT RESULT Users can browse and compare sampled palettes, inspect similarity, and trace results back to source photographs. The pipeline reproducibly generates and compares five-color palettes. LIMITATION Evidence is still based on classroom; samples are small and uneven, and five-color summaries may miss salient color accents. NEXT VALIDATION Structured task testing with intended users, plus parameter-sensitivity and human-similarity comparison.Owners: Interaction & Evaluation; Data Visualization & Prototyping.

Team roles and credits Tingyi Xiao: poster and visual communication. Jiaxin Wang: report/documentation. Andi Wan: demo video and interaction presentation. Data, code, visual, and AI-use attribution Photographs and data are team-produced. Visuals and project decisions are team-authored. AI assisted website prototyping and implementation; the team reviewed and verified the final code and outputs.

Professional reference: CHI 2026 Interactive Demos
https://chi2026.acm.org/authors/interactive-demos/
TactorBots paper and video: https://doi.org/10.1145/3544549.3583897 | https://www.youtube.com/watch?v=mGvSld62Mhk
CHI 2026 Posters: https://chi2026.acm.org/authors/posters/